

Supplemental Figures:

Figure S1

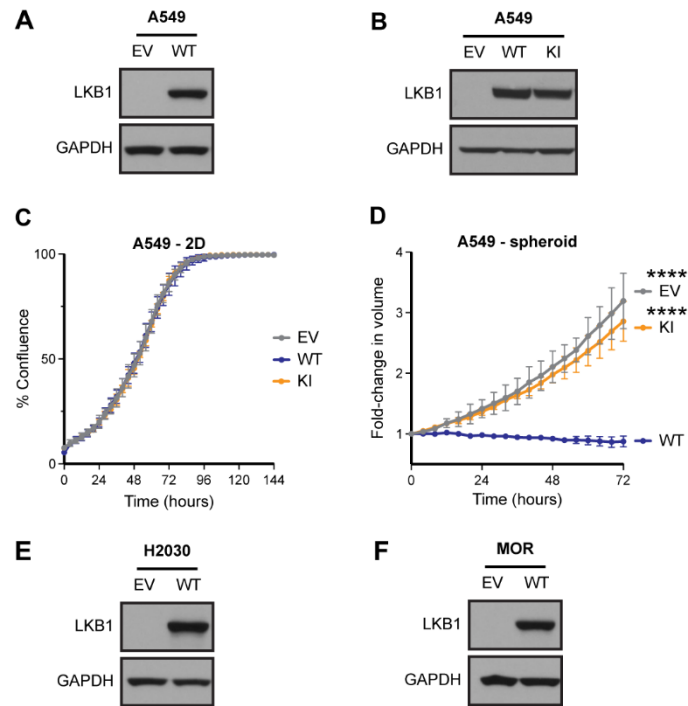


Figure S1: Expression of WT or kinase-inactive (KI) *LKB1* in *LKB1*-null LUAD lines. **(A, B)** *LKB1* protein levels in EV, WT, and KI cells (derived from A549). **(C)** Growth of A549 EV, WT, and KI cells in 2D culture. **(D)** Growth of A549 EV, WT, and KI cells in spheroid culture. **(E, F)** *LKB1* protein levels in EV and WT cells derived from H2030 and MOR LUAD lines.

Figure S2

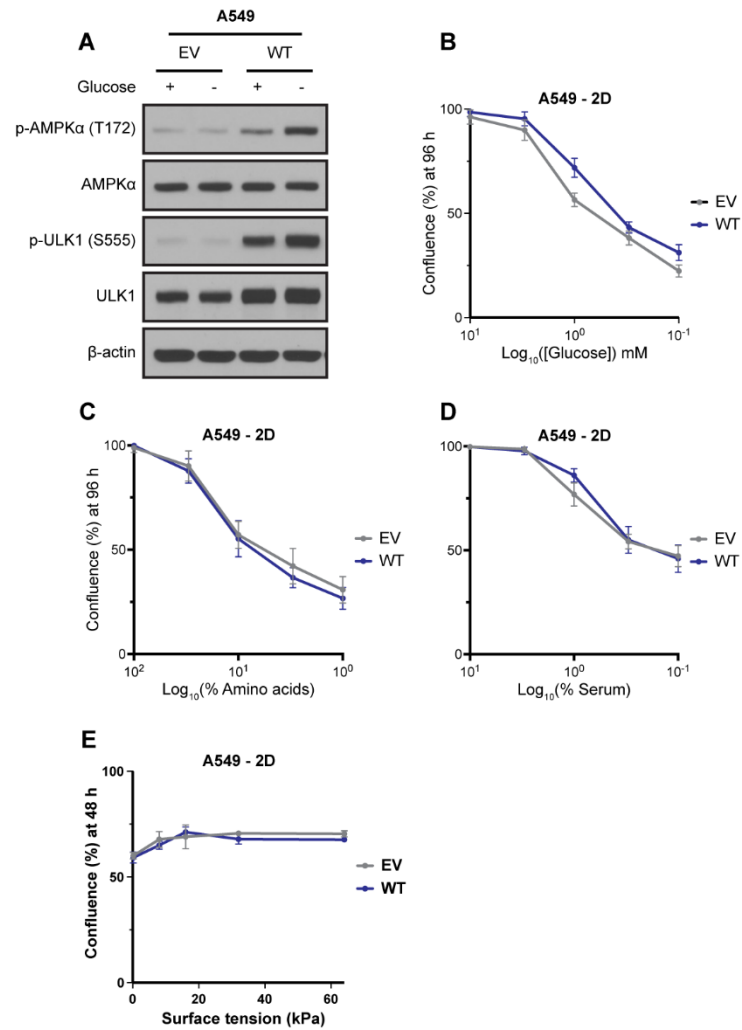


Figure S2: Nutrient deprivation, serum deprivation, or alterations in cellular surface tension do not recapitulate the tumor suppressive activity of LKB1 in 2D culture. **(A)** Levels of phosphorylated and total proteins in the LKB1-AMPK-ULK1 pathway from cells cultured in 2D, in media with or without glucose. **(B-E)** Growth of A549 EV and WT cells in 2D culture under variable levels of glucose **(B)**, amino acids **(C)**, serum **(D)**, or surface tension **(E)**. Error bars indicate standard deviation. $n = 3-6$.

Figure S3

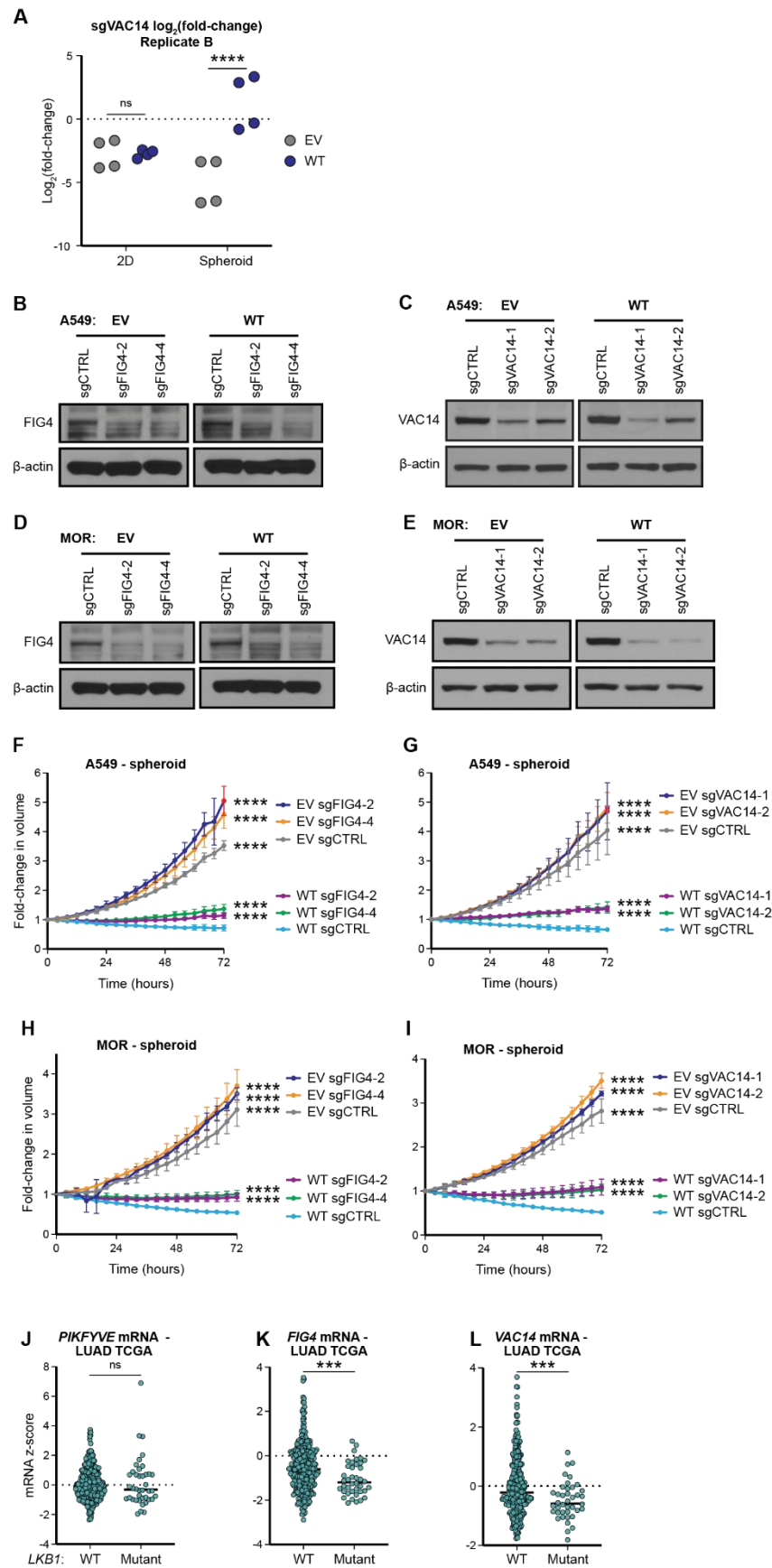


Figure S3: Validation of the genome-wide CRISPR screens. **(A)** Log₂(fold-change) values of individual sgRNAs corresponding to *VAC14* in EV and WT cells – grown in 2D or spheroid culture – from a second replicate of the CRISPR screens. **(B-E)** FIG4 and VAC4 protein levels in EV and WT cells (derived from A549 or MOR lines) containing sgCTRL, or sgRNAs against *FIG4* or *VAC14*. **(F-I)** Spheroid growth of EV and WT cells (derived from A549 or MOR lines) containing sgCTRL, or sgRNAs against *FIG4* or *VAC14*. Error bars indicate standard deviation. n = 3-6. **(J-L)** *PIKFYVE*, *FIG4*, and *VAC14* mRNA levels in *LKB1*-mutant and -WT tumors from the LUAD TCGA dataset. *** and **** indicate p<0.0002 and 0.0001. ns, non-significant. For the spheroid growth assays, p-values indicate pairwise statistical comparisons to “WT sgCTRL.”

Figure S4

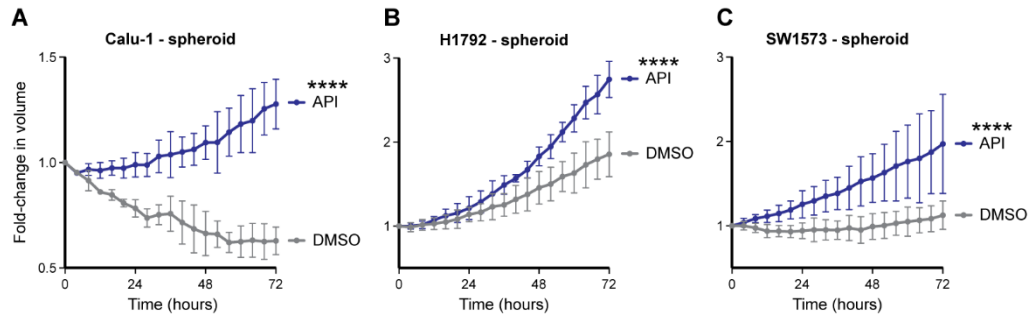


Figure S4: Inhibiting PIKFYVE promotes the growth of LUAD lines with endogenous WT *LKB1* as spheroids. **(A-C)** Spheroid growth of LUAD lines with endogenous WT *LKB1* (Calu-1 **(A)**, H1792 **(B)**, and SW1573 **(C)**) in the presence of DMSO or API. Error bars indicate standard deviation. n = 3-6. **** indicates p < 0.0001.

Figure S5

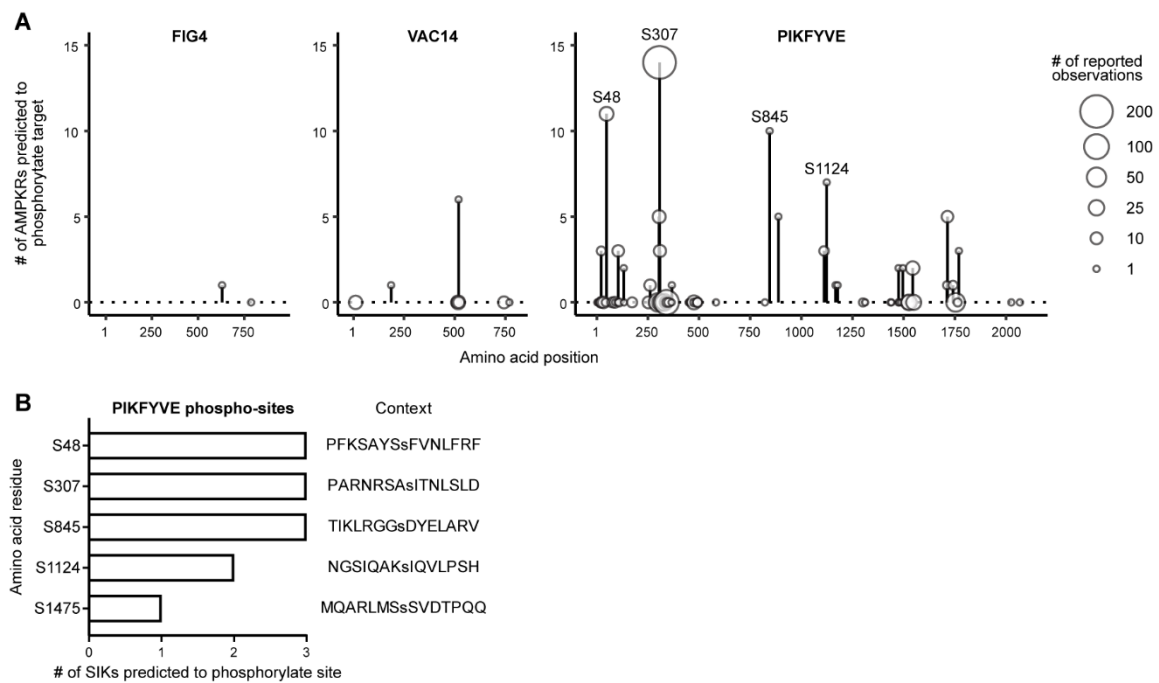


Figure S5: The SIKs are predicted to phosphorylate PIKFYVE. **(A)** The number of AMPKRs predicted to phosphorylate serine and threonine residues within the PIKFYVE complex. The size of each circle indicates the number of times a phosphorylated site has been publicly reported. **(B)** Amino acid residues on PIKFYVE that are predicted to be phosphorylated by the SIKs.

Figure S6

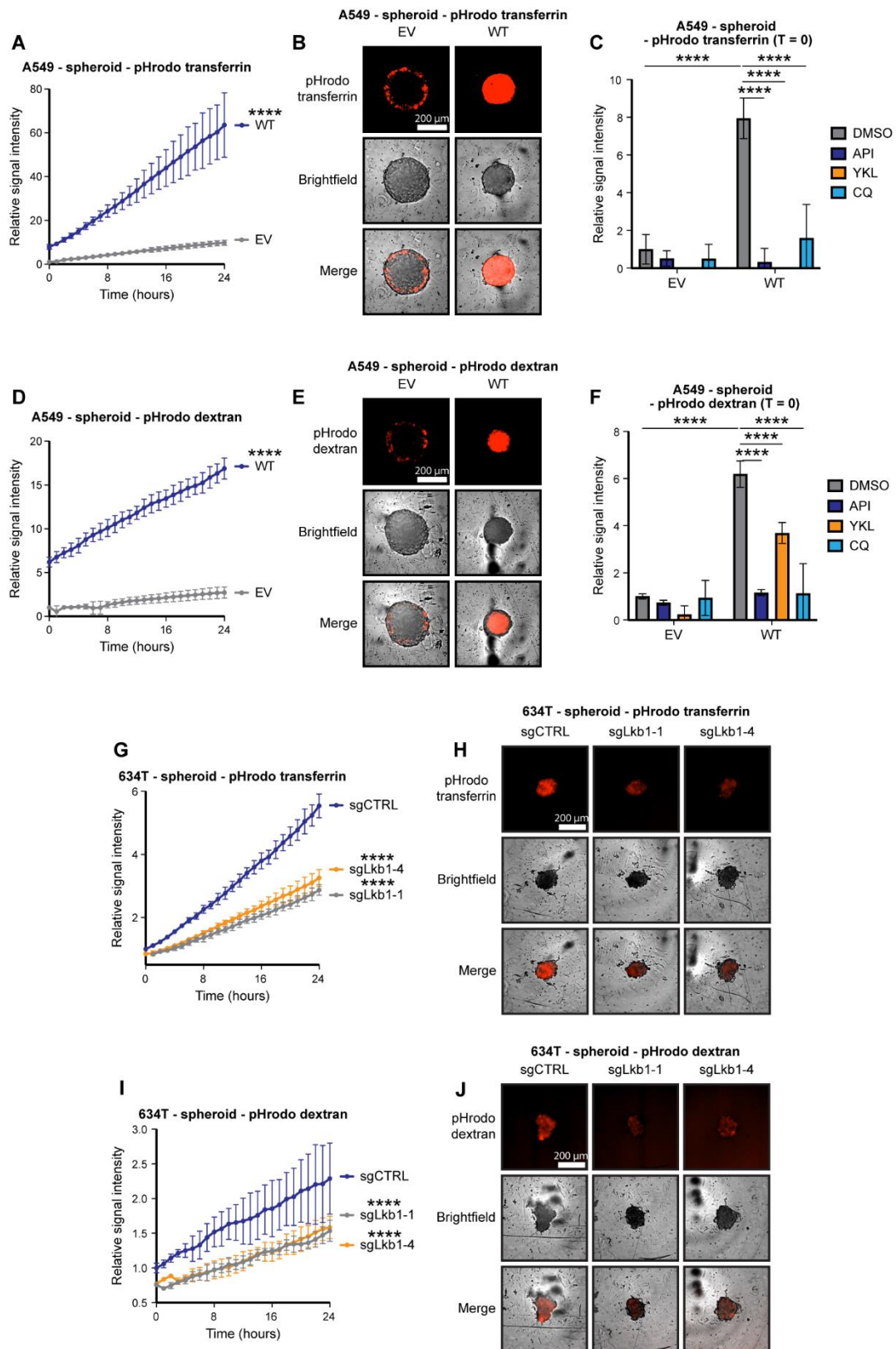


Figure S6: LKB1 promotes the endocytosis of transferrin and extracellular dextran through PIKFYVE. **(A)** Relative signal intensity of A549 EV and WT spheroids treated with a pH-sensitive reporter of the internalization of transferrin (pHrodo transferrin). **(B)** Representative images of EV and WT spheroids treated with pHrodo transferrin. **(C)** Relative signal intensity of EV and WT spheroids treated with DMSO, API, YLK, or CQ and pHrodo transferrin. **(D)** Relative signal intensity of A549 EV and WT spheroids treated with a pH-sensitive reporter of the endocytosis of dextran (pHrodo dextran). **(E)** Representative images of EV and WT spheroids treated with pHrodo dextran. **(F)** Relative signal intensity of EV and WT spheroids treated with DMSO, API, YLK, or CQ and pHrodo dextran. **(G-J)** Relative signal intensity and representative images of 634T spheroids containing sgCTRL or sgLkb1 and treated with pHrodo transferrin **(G, H)** or pHrodo dextran **(I, J)**. Error bars indicate standard deviation. n = 3-4. **** indicates $p < 0.0001$. For experiments with 634T cells, p-values indicate pairwise statistical comparisons to “sgCTRL.”

Figure S7

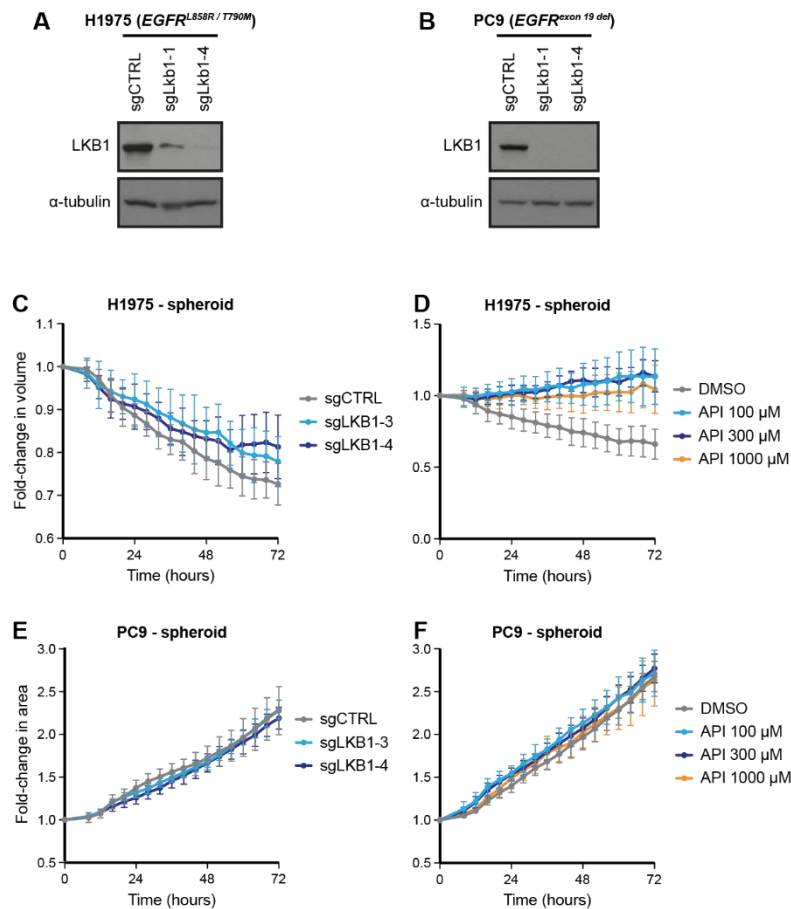


Figure S7: Neither loss of LKB1 nor inhibition of PIKFYVE impairs the growth of *EGFR*-mutant LUAD lines in spheroid culture. **(A, B)** LKB1 protein levels in *EGFR*-mutant LUAD lines (H1975 and PC9) that were transfected with a control sgRNA (sgCTRL) or sgRNAs against *LKB1*. **(C-F)** Spheroid growth of H1975 or PC9 cells with loss of LKB1 **(C, E)** or in the presence of various concentrations of API **(D, F)**. Error bars indicate standard deviation. n = 3-6.